

CS201 Midterm 2

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Abstract—Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures including arrays, (iv) you can use any function given in the data structure, (v) you can not modify the data structure contents.

I. QUESTION (QUEUE)

Write the method for array implementation

```
void cutPaste(Queue dest, int p, int q)
```

which cuts all the elements between indexes p and q from the original queue and inserts at the end to the dest queue. You are not allowed to use enqueue, dequeue, isEmpty functions. You can assume the destination queue has enough space for insertion. Your method should run in $\mathcal{O}(N)$ time.

II. QUESTION (QUEUE)

Write the method

```
Queue[] divideQueue(int k)
```

which constructs an array of list based queues by dividing the original queue into k equal parts. The first, second, ..., k 'th element of the original queue will be the first element of the first, second, ..., k 'th output queues, etc. The elements of the output queues should be recreated (not copied from the original queue). You are not allowed to use enqueue, dequeue, isEmpty functions. You should solve the question for list implementation.

III. QUESTION (BINARY SEARCH TREE)

Write the recursive method

```
int sumOfNodesBetween(int p, int q)
```

in `TreeNode` class which returns the sum of the keys between p and q in the tree. Your algorithm should

run in $\mathcal{O}(\log N + K)$, where K is the number of nodes which have value larger than p and less than q in the tree.

IV. QUESTION (BINARY SEARCH TREE)

Write the recursive method

```
int[] collectNodes()
```

in `TreeNode` class, which collects all values in all nodes in the tree in a sorted manner. You are not allowed to use any tree methods.

V. QUESTION (HASHING)

Write the static method in `Hash` class

```
boolean sumOfTwoK(int[] array, int K)
```

that takes an array of integers as a parameter and returns true if the sum of two elements in the array is K . Your method should run in $\mathcal{O}(N)$ time. Do not use any external data structures or arrays except the external hash table.

VI. QUESTION (HASHING)

Write the method in `Hash` class linked list implementation

```
boolean isValid()
```

that checks if the hash table is valid or not. An hash table is invalid if it contains the same number twice. Your method should run in $\mathcal{O}(N)$ time. Do not use external data structures or hash tables.